MAVIONIX INDIA PRIVATE LIMITED

WEB DEVELOPMENT INTERNSHIP – TECHNICAL ASSIGNMENT

Project: AI-Powered CRM – Intelligent Customer Relationship Management System

Duration: 7 Days
Difficulty: Intermediate
Submission Type: GitHub Repository + Live Deployment + Project Documentation

1. Project Overview

Develop a modern, responsive, AI-powered Customer Relationship Management (CRM) web application that helps businesses manage leads, track customer interactions, monitor sales activities, and generate AI-powered insights.

The objective is to build more than a traditional CRM dashboard. The application should use AI to help users understand their leads, identify potential opportunities, and make better decisions.

The application should have a clean, professional interface with a modern SaaS-style design.

2. Core Features and Requirements

A. Dashboard

Create an interactive dashboard displaying:

- Total leads
- New leads
- Converted customers
- Pending follow-ups
- Lead conversion rate
- Recent customer activities

Include charts or visual representations to show lead distribution and sales performance.

B. Lead Management System

Develop a complete lead management module.

Users should be able to:

- Add new leads.
- Edit and delete lead information.
- View individual lead profiles.
- Search leads by name, email, or company.
- Filter leads based on their status.
- Assign lead priorities.

Each lead should contain:

- Full name
- Email address
- Phone number
- Company name
- Lead source
- Lead status
- Priority
- Notes
- Creation date

Suggested lead statuses:

- New
- Contacted
- Interested
- Negotiation
- Converted
- Lost

C. Sales Pipeline

Create a visual sales pipeline using a Kanban-style layout.

Display leads under their respective stages.

Users should be able to move leads between stages using drag-and-drop functionality or an alternative stage-update interaction.

The interface should update the lead status accordingly.

D. AI-Powered Lead Intelligence

This is the most important feature of the assignment.

Integrate an AI feature that can analyze individual lead information and provide:

1. Lead Summary: Generate a short summary of the customer's profile and requirements.
2. Lead Scoring: Assign a score between 1 and 100 based on the available lead information.
3. Conversion Probability: Provide an estimated likelihood of conversion.
4. Next Best Action: Suggest what the sales representative should do next.

5. Follow-up Message: Generate a personalized professional follow-up email or message.

Example:

A lead has shown interest in a company's software product but has not responded to the last two follow-up messages.

The AI should analyze the available information and suggest an appropriate next action.

Candidates may use an LLM API such as Gemini or OpenAI. If API access is unavailable, a clearly documented mock AI response is acceptable for demonstration purposes.

Do not expose API keys in frontend code or commit them to GitHub.

E. Customer Interaction History

Create an activity timeline for each lead.

It should record:

- Calls
- Emails
- Meetings
- Notes
- Follow-up activities

Users should be able to add new interaction records and view previous activities.

F. Follow-up Management

Develop a basic follow-up management module.

Users should be able to:

- Create follow-up reminders.
- Set follow-up dates.
- View upcoming follow-ups.
- Mark follow-ups as completed.
- Identify overdue follow-ups.

3. Suggested Technology Stack

Candidates are encouraged to use modern technologies.

Frontend:

- Next.js or React.js
- TypeScript
- Tailwind CSS

- Shadcn/UI or another modern component library
- Recharts for dashboard visualizations

Backend:

- Next.js API Routes or Node.js with Express
- REST APIs

Database:

- Supabase with PostgreSQL
- Prisma ORM (optional)

AI Integration:

- Gemini API or OpenAI API
- Structured AI responses using JSON where possible

Additional Libraries:

- TanStack Query
- React Hook Form
- Zod
- dnd-kit for drag-and-drop functionality

The suggested stack is not mandatory. Candidates may use equivalent technologies if they can explain their implementation.

4. UI/UX Expectations

The application should have:

- A modern SaaS dashboard.
- Clean sidebar navigation.
- Responsive layouts.
- Consistent typography and spacing.
- Interactive charts and tables.
- Proper loading and error states.
- Useful empty states.
- Smooth interactions and transitions.

The design should look like a real business product rather than a basic college project.

5. Additional Challenge (Optional)

Candidates who want to demonstrate additional skills may implement one or more of the following:

- AI chatbot for querying CRM data.
- Natural-language lead search.
- AI-generated daily sales summaries.
- CSV import and export.
- Role-based access for Admin and Sales Representatives.
- Email integration or automated follow-up workflows.

These features are optional and will be considered additional contributions.`

6. Evaluation Criteria

Evaluation Area	Weightage
UI/UX and Visual Design	15%
Frontend Development	20%
Backend and Database Integration	15%
AI Integration and Functionality	20%
Core CRM Features	15%
Code Quality and Project Structure	10%
Documentation and Deployment	5%
Total	100%

7. Submission Requirements

Candidates must submit:

1. Live Deployment Link: A working hosted version of the application.
2. GitHub Repository: Complete source code with a proper folder structure.
3. README File: Include project overview, technologies used, setup instructions, and implemented features.
4. Demo Video: A short 2–3 minute walkthrough explaining the application and demonstrating its main features.
5. AI Implementation Details: Explain which AI model/API was used and how the AI functionality works. If mock responses were used, clearly mention this.

Ensure that the GitHub repository is accessible to the review team.

8. Important Instructions

- The assignment must be completed within 7 days of receiving it.
- Candidates may refer to documentation and use AI development tools.
- However, candidates must understand and be able to explain their code and implementation.
- Do not submit copied projects or existing CRM templates without meaningful customization.
- All major features should be functional.
- Keep the code organized, readable, and maintainable.
- Do not publish API keys, passwords, or confidential information.

9. Final Objective

We are looking for candidates who can think beyond basic website development and build a functional business application using modern web technologies and AI capabilities.

The focus is on practical implementation, problem-solving, creativity, technical understanding, and the ability to turn an idea into a working product.

We look forward to reviewing your work.

HR Team
MaVionix India Private Limited

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